

High-performance thin-film lithium niobate electro-optic modulator based on etching slot and ultrathin silicon film

YI WANG,¹ YIN XU,^{1,2}  BO ZHANG,^{1,2} YUE DONG,^{1,2}  AND YI NI^{1,2,*}

¹Department of Electronic Engineering, School of IoT Engineering, Jiangnan University, Wuxi 214122, China

²Institute of Advanced Technology, Jiangnan University, Wuxi 214122, China

*8073110160@jiangnan.edu.cn

Received 29 November 2022; revised 25 January 2023; accepted 26 January 2023; posted 27 January 2023; published 24 February 2023

An electro-optic modulator (EOM) is an indispensable component to connect the electric and optical fields. Here, we propose a high-performance, thin-film lithium niobate-based EOM, where the modulation waveguide is formed by an etching slot on the lithium niobate film and the deposit of an ultrathin silicon film in the slot region. Therefore, a small mode size and high mode energy can be simultaneously achieved in the LN region with a high EO coefficient, which will be beneficial to increase the EO overlap and gradually decrease in the mode size. Further, we employed a waveguide structure to construct a typical Mach-Zehnder interference-type EOM. According to the requirements of high-speed traveling wave modulation, we conduct the index matching, impedance matching, and low-loss operation. From the results, the key half-wave voltage length product and 3 dB modulation bandwidth are, respectively, 1.45 V cm and 119 GHz in a modulation length of 4 mm. Moreover, a larger 3 dB bandwidth also can be achieved by shortening the modulation length. Therefore, we believe the proposed waveguide structure and EOM will provide new ways to enhance the performance of LNOI-based EOMs. © 2023 Optica Publishing Group

<https://doi.org/10.1364/AO.482202>

1. INTRODUCTION

An on-chip electro-optic modulator (EOM), a vital bridge that connects the electric and optical fields on the same chip, is an indispensable component for on-chip optical communications based on photonic integrated circuits (PICs) [1–5]. Faced with the rapid increase in the on-chip transmission capacity and bandwidth, high-performance PICs must be developed, where the EOM gradually becomes a performance bottleneck since the low-speed and narrow bandwidth modulation strongly limits and even stops the transmission performance enhancement of PICs [1,6–8]. Therefore, high-performance EOM is our goal.

To enhance the EOM performance, the key issue is to choose optimum modulation materials that have a high EO coefficient and can be easily combined with conventional waveguide structures. Until now, the following materials have been used for EOMs: indium phosphide [9,10], electro-optic polymers [11,12], 2D materials (e.g., graphene) [13,14], doped silicon [15,16], and lithium niobate (LN) [17,18]. Among them, LN is a widely used material for the EOM since LN has a wide transparent window (from 400 nm to 5 μ m) and a large EO. It also has piezoelectric, photoelastic, and nonlinear coefficients together with quite stable physical and chemical properties [19,20]. However, an LN-based EOM also has an obvious drawback for integrated photonics; that is, it has a large device size that can barely be integrated on the chip with other components.

The main reason is that the fundamental waveguide structure in the LN-based EOM is a low refractive-index-contrast waveguide formed by titanium indiffusion or proton exchange in the bulk LN [19,20]. Recently, a single-crystalline thin-film LN-on-insulator (LNOI) has been successfully fabricated using crystal ion slicing and the LNOI wafer has become an important commercial product [21,22], similar to a mature silicon-on-insulator (SOI) wafer. With a relatively high refractive-index contrast for the LNOI wafer ($\Delta n > 0.7$), the mode confinement of an LNOI-based waveguide is clearly enhanced compared to a traditional bulk LN-based waveguide and the typical single-mode LNOI waveguide width is reduced to ~ 1 μ m, which could well support the development of LNOI-based integrated photonics [21–24]. Meanwhile, the excellent material properties of LN can be well kept for the LNOI wafer practically without obvious deterioration. So, various photonic devices based on the LNOI platform have been proposed and demonstrated lately, especially for the EOM [25–27].

The current reported LNOI-based EOM has the obvious advantages of a low half-wave-voltage-length product ($V_\pi L$), a large modulation bandwidth, and a compact size compared to a conventionally used LN-based EOM. For example, by etching the LNOI wafer to form the ridge waveguide and further depositing the modulation electrodes on both of its sides [28–32], a typical EOM can be obtained where the $V_\pi L$ and

3 dB modulation bandwidth can be reduced to 1.75 V cm and increased to 40 GHz, respectively [28]. Considering the etching difficulties of LNOI film at present, an etchless scheme was also proposed [33,34]. By patterning other waveguide materials (e.g., polymer [33], Si₃N₄ [35–37], and Si [38,39]) atop or below the etchless LNOI film, a waveguide structure can be formed that will confine the optical mode. If we further add the modulation electrodes on the LNOI film to apply a radio frequency (RF) field on the optical mode, obvious EO modulation phenomenon can be observed and the obtained $V_\pi L$ and 3 dB modulation bandwidth are 2.24 V cm [35], 6.7 V cm [38], ~30 GHz [35], and 106 GHz [38], respectively. Note that these are currently the most commonly used modulation waveguide structures based on the LNOI platform. To take full advantage of the material properties of LN, however, a new modulation waveguide structure should be further developed, which could help achieve a lower $V_\pi L$, larger bandwidth, as well as a reduced modulation length.

In this paper, we propose a new heterogeneously integrated EOM based on the LNOI platform. The vital optical waveguide structure is formed by etching the LNOI film and then depositing an ultrathin silicon film in the etched region. Owing to the high refractive index of silicon compared to LN, the optical mode can be formed in a relatively small etched region where the mode energy is mainly confined in the LNOI film since the used silicon film is quite thin. Based on such a waveguide structure, the small mode size and high mode energy located in the LNOI film can be achieved simultaneously, which will be very beneficial to enhance the EO overlap in a relatively small size. To study the modulation performance of the proposed optical waveguide, we use the proposed waveguide structure to build a Mach–Zehnder interference (MZI) type modulator where the modulation electrodes are arranged in a conventional ground-signal-ground (G-S-G) configuration. Further, the device design and optimization are aimed at index matching, impedance matching, and low-loss operation. From results, the obtained $V_\pi L$ and 3 dB modulation bandwidth of the proposed EOM are 1.45 V cm and 119 GHz, respectively, which are clearly better than those of previous reports [28–30,36–38]. Therefore, we hope the proposed waveguide structure and corresponding EOM could offer an effective device scheme to promote the development of an LNOI-based EOM.

2. DEVICE STRUCTURE AND PRINCIPLE

Figure 1 shows the schematic of our proposed EOM on a 600 nm thick X-cut LNOI wafer. By etching a slot on the LNOI film and then depositing an ultrathin silicon film in the slot region, a new waveguide structure can be formed. The top width, depth, and slant angle of the etching slot are W , h_{slot} , and θ ($= 85^\circ$), respectively, where the thickness of the deposited silicon film is h_{Si} . Based on the clear refractive index difference between the silicon and LN, the optical mode can be formed for the proposed waveguide structure. Compared to the commonly used ridge LNOI waveguide and etchless LNOI waveguide [28–30,33–38], the proposed waveguide structure can simultaneously achieve the small mode size and high mode energy in the LNOI film. Further, the modulation electrodes are introduced atop the LNOI film in a G-S-G configuration

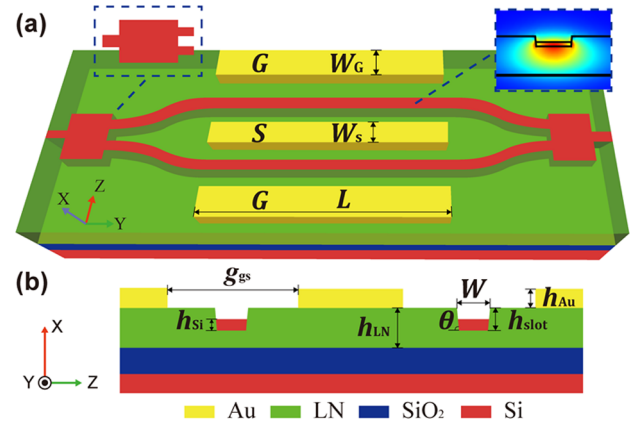


Fig. 1. (a) Schematic of the proposed EOM based on the LNOI platform. Insets show the structure of the power splitter and the zoomed-in view of the optical waveguide mode in the modulation region. (b) Cross-sectional view of the modulation region shown in (a). The material and structural parameters also are labeled.

around the waveguides. The electrode thickness is h_{Au} , and the gap width between the adjacent electrodes is g_{gs} , including the electrode length L . In addition, the electrode widths of the ground and signal electrodes are, respectively, W_G and W_S . So, the key modulation structure is generated, and we will analyze this structure in the following sections.

To demonstrate the performance of the proposed modulation structure, we conducted a detailed comparison between the conventionally used modulation structures and the present one based on the LNOI platform, where the optical mode, RF mode, and the key parameters (e.g., optical loss α_{op} , EO overlap factor Γ , and $V_\pi L$) are analyzed as shown in Fig. 2. $V_\pi L$ is defined as [39]

$$V_\pi L = \frac{\lambda g_{\text{gs}} n_{\text{eff}}}{2n_e^4 r_{33} \Gamma}, \quad (1)$$

where λ is the wavelength ($\lambda = 1.55 \mu\text{m}$ in here), n_{eff} is the optical effective mode index, n_e is the extraordinary refractive index of LN ($n_e = 2.14$ at $\lambda = 1550 \text{ nm}$), r_{33} is the electro-optic coefficient ($r_{33} \approx 31 \text{ pm/V}$ [21]), and Γ is the normalized overlap factor between the optical field and the microwave field within the LN region, defined as [39]

$$\Gamma = \frac{g_{\text{gs}}}{V} \frac{\iint_{\text{LN}} E_{\text{op}}^2(x, z) E_{\text{es}}(x, z) dx dz}{\iint_{\Omega} E_{\text{op}}^2(x, z) dx dz}, \quad (2)$$

where V is the applied voltage for the proposed modulator, $E_{\text{op}}(x, z)$ is the electric field distribution of the optical mode, $E_{\text{es}}(x, z)$ is the electrostatic field, and the integral domains of the numerator and the denominator are, respectively, the LN region and the whole cross section of the waveguide that is the half region illustrated in Fig. 1(b). To make a fair comparison, the gap g_{gs} width of the adjacent electrodes, and the thicknesses of LN film and silicon film are kept the same shown in Fig. 2. From Fig. 2, we can find that the ridge LNOI waveguide has the largest mode size, and the etchless LNOI waveguide has smaller mode size, but its mode energy should be more distributed in the LNOI region. By comparison, our proposed waveguide structure can achieve a relatively small mode size and the high mode

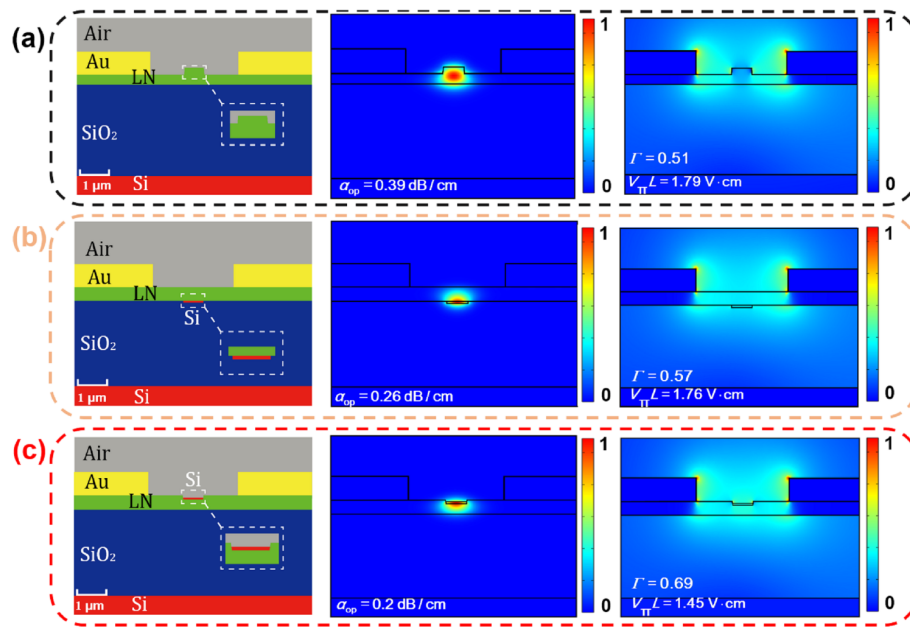


Fig. 2. Comparison of modulation structures, optical modes, and RF modes of different waveguides. (a) Ridge LNOI waveguide, (b) etchless LNOI waveguide, and (c) the proposed waveguide structure. The calculated parameters of optical loss, EO overlap factor, and $V_{\pi}L$ are also marked. The electrode gaps are set to the same value of 4 μm .

energy is confined in the LNOI waveguide due to the etched slot and deposited ultrathin silicon film. Based on these features, the modulation electrode could get closer to each other and the EO overlap factor could also be increased, contributing to the reduction of the key modulation parameter $V_{\pi}L$. According to the calculation results, the proposed modulation structure has the largest EO overlap factor and the achieved $V_{\pi}L$ is less than 1.5 V·cm, which is relatively good compared to previous reports [28–30,36–38]. Meanwhile, the optical loss of the proposed modulation structure is also relatively low (0.2 dB/cm), which could support the on-chip centimeter-level optical transmission. Here, such optical loss mainly comes from the metal absorption loss, while the scattering loss is not considered since it is dependent on the surface roughness of the etched slot in the LNOI wafer.

The working principle and process for the EOM based on the proposed waveguide structure are described next. As the input light enters into the waveguide, it first will be divided into two parts evenly via a power splitter and then it will enter into the interference arms. Based on the typical G-S-G configuration for the modulation electrodes, the EOM can work in a “push-pull” manner that could shorten the modulation length and enhance the EOM performance [21]. For the proposed modulation waveguide structure, it could make the optical mode and RF mode interact well with each other in the LNOI region. In addition, the mode size is also relatively small. As a result, the modulation performance could be effectively enhanced. Meanwhile, the introduced slot structure can also improve the optical group index suitably, which can match well with the RF mode index. After the modulation, two light beams will be combined together through light interference, where the accumulated phase difference between these two beams will directly affect the interference pattern. Finally, at the device output,

we found that the output spectrum will shift compared to if no modulation voltage was applied. Therefore, in the following analyses, we will study and find the optimum parameters of the proposed modulation waveguide and electrodes to achieve better modulation performance compared to previously reported EOMs based on the LNOI platform [28–30,36–38].

3. RESULTS AND DISCUSSION

For the high-speed EOM, the device design and optimization should be aimed at achieving index matching, impedance matching, and low-loss operation. With these requirements, we analyzed and determined the structural parameters based on COMSOL Multiphysics the simulation tool [40]. Figure 3(a) shows the calculated minimum value of the top width ($W_{\min}$) of the etching slot for the proposed modulation waveguide as a function of the electrode gap width g_{gs} under the condition of an optical loss less than 0.2 dB/cm, where several typical thicknesses (h_{slot} , h_{Si}) for the etching slot and silicon film are also considered. From Fig. 3(a), we clearly found that $W_{\min}$ decreases gradually with an increase in g_{gs} . Therefore, if we want to achieve small $W_{\min}$, the electrode gap width must be enlarged, but at the cost of increased $V_{\pi}L$, which is further analyzed in Fig. 3(b). The top width of the etching slot is set to $W_{\min}$, which was obtained from Fig. 3(a). In addition, the minimum electrode gap was set to 4 μm by considering the metal absorption loss and the index matching. Figure 3(b) illustrates the calculated $V_{\pi}L$ versus the electrode gap width g_{gs} , where $h_{\text{slot}} = 0.15, 0.25 \mu\text{m}$, and $h_{\text{Si}} = 0.07, 0.09 \mu\text{m}$ are considered. Note that $V_{\pi}L$ nearly increases with an increase in g_{gs} since a larger gap width will require higher half-wave-voltage (V_{π}) to drive the modulator with the same modulation length. Under such a condition, to achieve a low $V_{\pi}L$, the electrode gap width should

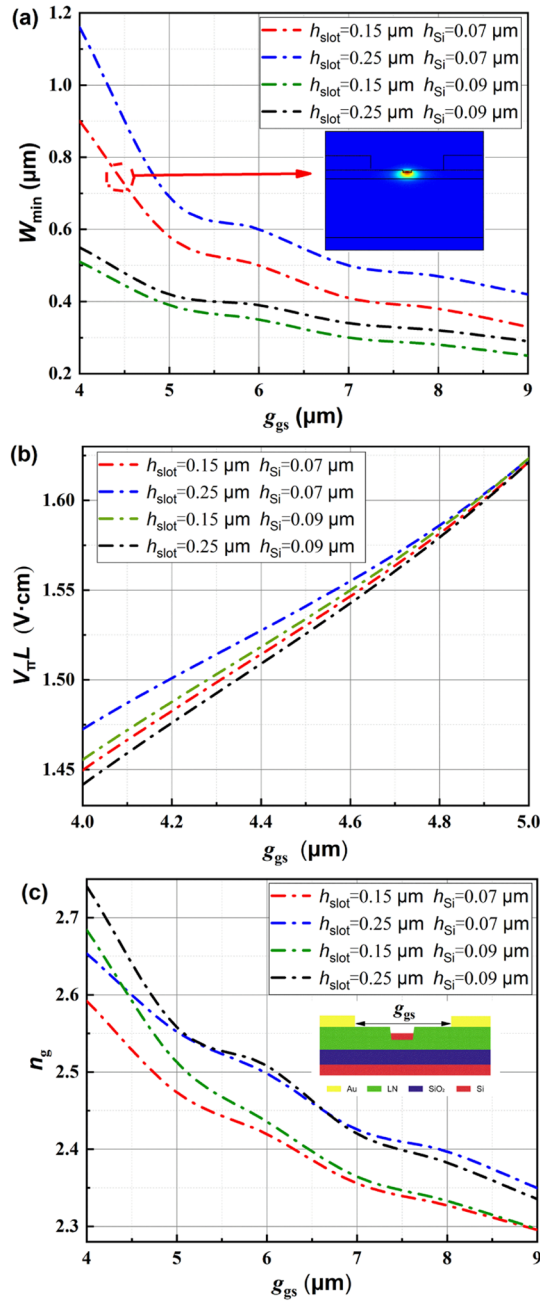


Fig. 3. (a) Minimum value of the top width ($W_{\min}$) of the etching slot for the proposed modulation waveguide versus the electrode gap width g_{gs} . Inset shows the electric field distribution of the optical mode. (b) $V_{\pi}L$ versus the electrode gap width g_{gs} . (c) Optical group index (n_g) versus the electrode gap width g_{gs} . Inset shows the calculated device structure. The thicknesses of etching slot and silicon film are $h_{\text{slot}} = 0.15, 0.25 \mu\text{m}$, and $h_{\text{Si}} = 0.07, 0.09 \mu\text{m}$, respectively, which has been considered in the calculations.

be reduced, which requires a small mode size to guarantee the low-loss operation. Moreover, we also analyzed the optical group index (n_g) as a function of the electrode gap width g_{gs} , as shown in Fig. 3(c), where $h_{\text{slot}} = 0.15, 0.25 \mu\text{m}$, and $h_{\text{Si}} = 0.07, 0.09 \mu\text{m}$ are considered. The calculated n_g reveals a decreasing trend with an increase in g_{gs} . To create a good match between the optical mode index and the RF mode index, n_g should be

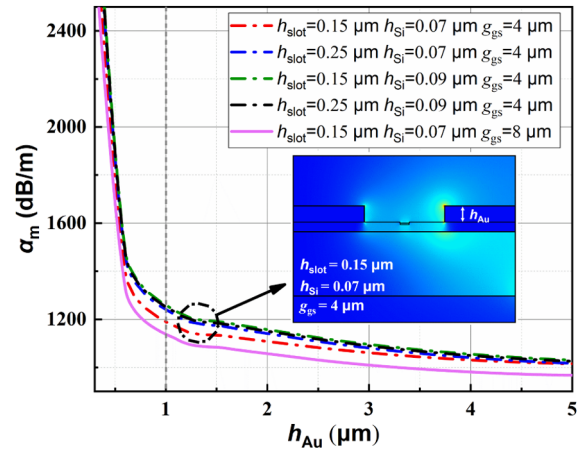


Fig. 4. RF loss α_m versus the electrode thickness h_{Au} , where the thicknesses of the etching slot and silicon film are considered ($h_{\text{slot}} = 0.15, 0.25 \mu\text{m}$, $h_{\text{Si}} = 0.07, 0.09 \mu\text{m}$) together with the electrode gap width ($g_{gs} = 4, 8 \mu\text{m}$). Inset shows the RF field pattern with $h_{\text{slot}} = 0.15 \mu\text{m}$, $h_{\text{Si}} = 0.07 \mu\text{m}$, and $g_{gs} = 4 \mu\text{m}$.

increased, corresponding to the small value of g_{gs} . In addition, we also conducted an RF analysis of the proposed modulation structure, and the RF mode field can also be obtained with the help of COMSOL Multiphysics software [40].

Figure 4 shows the RF loss α_m of the proposed structure as a function of its electrode thickness h_{Au} , where the RF loss is closely related to the imaginary part of the RF mode effective index [29]. Note that the RF loss decreases with an increase in the electrode thickness h_{Au} . When $h_{\text{Au}} < 1 \mu\text{m}$, the RF loss decreases quickly with an increase in h_{Au} . In contrast, when $h_{\text{Au}} > 1 \mu\text{m}$, the RF loss decreases quite slowly and gradually tends to become stable with a further increase in h_{Au} . Thus, a thick electrode will introduce low RF loss and could support relatively a long distance for the on-chip transmission, which means the length of the modulation electrode can be increased, leading to a reduction in the driving voltage V_{π} . However, the electrode thickness is also related to the impedance, and we should carefully design the electrode thickness, which would further affect the requirement for impedance matching. Therefore, by both considering the RF loss and the impedance matching, we set the electrode thickness as $h_{\text{Au}} = 1 \mu\text{m}$ under the following analyses.

Next, we conducted the impedance matching, which mainly focused on the metal electrode. According to the requirement of impedance matching, the characteristic impedance of the proposed EOM should be designed to be 50Ω . With this requirement, we optimized the signal electrode width W_s and the electrode gap width g_{gs} , where $h_{\text{slot}} = 0.15, 0.25 \mu\text{m}$, and $h_{\text{Si}} = 0.07, 0.09 \mu\text{m}$ are also considered, as shown in Fig. 5. From results, we found that W_s must be increased as g_{gs} becomes large, which means a larger electrode gap width that requires a larger electrode width to satisfying the same impedance. Meanwhile, this relationship is also kept for different thicknesses of the etching slot and silicon film. Therefore, we can conclude that the electrode gap width and signal electrode width are closely related. When the electrode gap width is determined, the signal electrode width can also be fixed from Fig. 5, which is based on the requirement for

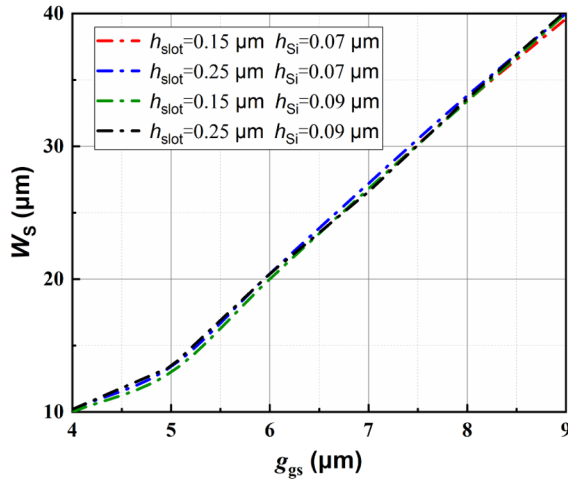


Fig. 5. Impedance matching analysis. Signal electrode width W_s as a function of the electrode gap width g_{gs} , when the characteristic impedance of the proposed EOM is set to 50Ω . The thicknesses of etching slot and silicon film are respectively, $h_{\text{slot}} = 0.15, 0.25 \mu\text{m}$ and $h_{\text{Si}} = 0.07, 0.09 \mu\text{m}$.

impedance matching. Figure 6 shows the index matching between the RF mode effective index n_{RF} and the optical group index n_g as a function of the electrode gap width g_{gs} , where $h_{\text{slot}} = 0.15, 0.25 \mu\text{m}$, and $h_{\text{Si}} = 0.07, 0.09 \mu\text{m}$ are also considered. From Fig. 6, with an increase in g_{gs} , n_{RF} and n_g reveal opposite changing behavior, and their crossing points are within the range from $g_{gs} = 4$ to $5 \mu\text{m}$, corresponding to the index-matching positions. According to the calculation results, the index-matching positions are near $g_{gs} = 4 \mu\text{m}$, $4.4 \mu\text{m}$, and $4.7 \mu\text{m}$, respectively, which are also marked as shadow regions in Fig. 6. The corresponding thicknesses of the etching slot and silicon film are $h_{\text{slot}} = 0.15 \mu\text{m}$, $h_{\text{Si}} = 0.07 \mu\text{m}$ ($g_{gs} = 4 \mu\text{m}$), $h_{\text{slot}} = 0.25 \mu\text{m}$, $h_{\text{Si}} = 0.07 \mu\text{m}$, $h_{\text{slot}} = 0.15 \mu\text{m}$, $h_{\text{Si}} = 0.09 \mu\text{m}$ ($g_{gs} = 4.4 \mu\text{m}$), and $h_{\text{slot}} = 0.25 \mu\text{m}$, $h_{\text{Si}} = 0.09 \mu\text{m}$ ($g_{gs} = 4.7 \mu\text{m}$), respectively. Therefore, based on the results of impedance matching, index matching, and low-loss operation, we further studied the modulation bandwidth of the proposed modulator.

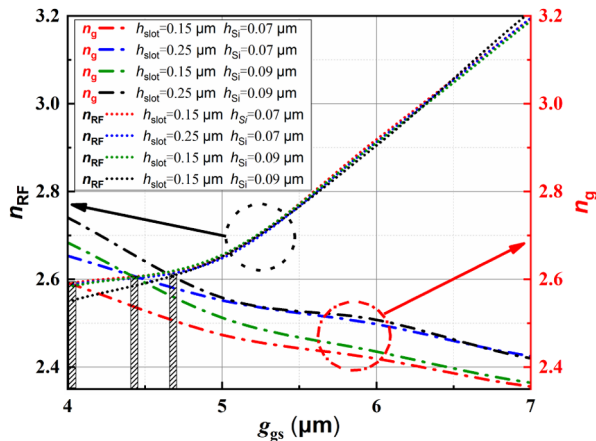


Fig. 6. Index matching analysis. RF mode effective index n_{RF} and optical group index n_g as a function of the electrode gap width g_{gs} , where $h_{\text{slot}} = 0.15, 0.25 \mu\text{m}$, and $h_{\text{Si}} = 0.07, 0.09 \mu\text{m}$ are included. The shadow regions are the index-matching regions.

To determine the modulation bandwidth, the frequency response of the proposed modulator should be studied. The frequency response $m(\omega)$ of the EOM can be expressed as [29]

$$m(\omega) = \left| \frac{2Z_{\text{in}}}{Z_{\text{in}} + Z_L} \frac{(Z_L + Z_0)F_+ + (Z_L - Z_0)F_-}{(Z_L + Z_0)e^{\gamma_m L} + (Z_L - Z_0)e^{-\gamma_m L}} \right|^2, \quad (3)$$

$$Z_{\text{in}} = Z_0 \frac{Z_L + Z_0 \tanh(\gamma_m L)}{Z_0 + Z_L \tanh(\gamma_m L)}, \quad (4)$$

$$F_{\pm} = \frac{1 - e^{\pm \gamma_m L - j \frac{\omega}{c_0} n_g L}}{\pm \gamma_m L - j \frac{\omega}{c_0} n_g L}, \quad (5)$$

$$\gamma_m = \alpha_m + j \frac{\omega}{c_0} n_{\text{RF}}, \quad (6)$$

where Z_{in} is the input impedance of the EOM, Z_L is the load and source impedance (50Ω used in here), Z_0 is the characteristic impedance, $F_{\pm}$ is the forward/backward propagating waves, γ_m is the complex RF mode propagation constant (α_m being the RF propagation loss, n_{RF} being the RF mode effective index), ω is the RF frequency, c_0 is the light speed in the vacuum, and L is the modulation length. Based on the frequency response expression, we can calculate the modulation performance of the proposed EOM. Figure 7 shows the EO response of the proposed device under the index-matching positions ($g_{gs} = 4 \mu\text{m}$, $4.4 \mu\text{m}$, and $4.7 \mu\text{m}$), where the modulation length L is 4 mm . According to the calculation results, the obtained 3 dB EO bandwidths are 119, 123, and 126 GHz at the index-matching positions, where the impedance matchings are also satisfied at these positions. Therefore, through careful design of the structural parameters to satisfying the requirements of index matching, impedance matching, and low-loss operation ($< 0.2 \text{ dB/cm}$), the 3 dB EO bandwidth can be higher than 110 GHz, which is very promising for on-chip high-speed transmission. To achieve a higher 3 dB EO bandwidth, shortening the modulation length is the most efficient way; however, it will lead to an increase in the half-wave voltage V_{π} , corresponding to an enhanced energy consumption [21].

For device fabrication, its main fabrication process can be divided into three sections: fabricating the etching slot, depositing the silicon film, and adding the modulation electrodes. We started from a 600 nm thick X-cut LNOI wafer. First, the slot with a top width of $W = 900 \text{ nm}$ and a depth of $h_{\text{slot}} = 150 \text{ nm}$ was etched on the LNOI wafer, where the slant angle was set to 85° since an etching depth of only 150 nm currently can achieve such large slant angle [21]. Second, an ultrathin silicon film with a thickness of 70 nm was deposited on the LNOI wafer using the molecular beam epitaxy; then the redundant silicon film was removed, except in the slot region, where the silicon film was well deposited in the bottom of the etched slot. Third, the modulation electrodes were fabricated on the LNOI wafer near the waveguides based on the typical metal lift-off process [41]. Finally, we realized the designed EOM based on the new waveguide structure.

To better show the obtained device performance compared to some previous reports, Table 1 offers a detailed comparison. Table 1 shows that the proposed EOM has clear advantages

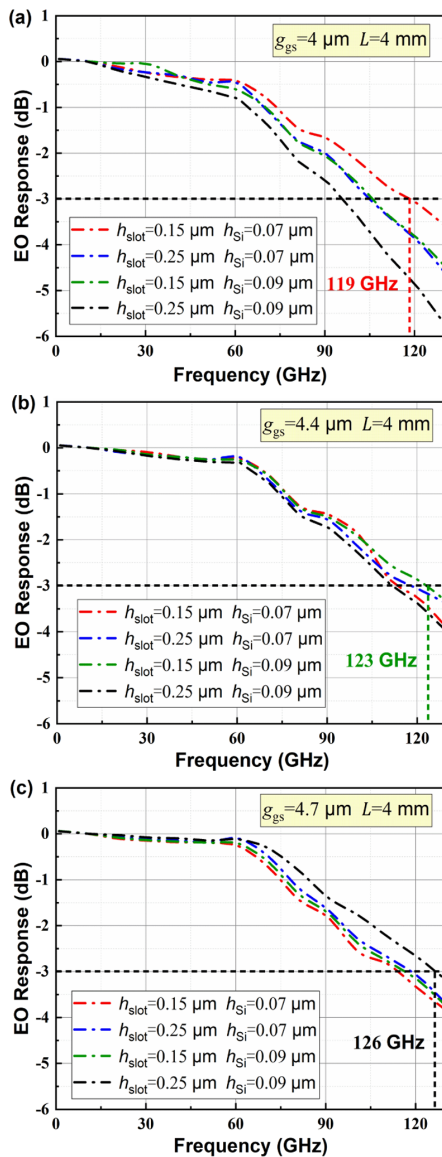


Fig. 7. Frequency response analysis. (a) $g_{gs} = 4$ mm, (b) $g_{gs} = 4.4$ mm, and (c) $g_{gs} = 4.7$ mm. The parameters of $h_{slot} = 0.15$, 0.25 mm and $h_{Si} = 0.07$, 0.09 mm are included. The modulation length is 4 mm.

in $V_{\pi}L$, 3 dB bandwidth, and modulation length, which are derived from the new waveguide structure. Therefore, with the features of a low $V_{\pi}L$ (1.45 V cm), a large 3 dB bandwidth (119 GHz), and a short modulation length (4 mm), we believe the proposed waveguide structure and related EOM could promote the development of LNOI-based EOMs and support on-chip high-speed transmission [25–30].

4. CONCLUSION

In conclusion, we proposed a high-performance LNOI-based EOM, where the key modulation waveguide is formed by etching a slot and depositing an ultrathin silicon film. Based on the refractive index difference between the ultrathin silicon film and LN, a small mode size and high mode energy can be generated in the LN region with a high EO coefficient. Both features will contribute to the enhancement of the EO overlap in a small size. Using such a waveguide, we further constructed an MZI-type EOM and the modulation electrodes were deposited near the waveguides in a G-S-G configuration. To achieve high-speed traveling wave modulation, we conducted the structural design and optimization to realize index matching, impedance matching, and the low loss operation. According to the calculation results, the key $V_{\pi}L$ and 3 dB modulation bandwidth are 1.45 V cm and 119 GHz, respectively, together with a modulation length of 4 mm, which are superior results to some previous reports. With these advantages, we believe the proposed waveguide structure and the corresponding EOM could offer new ideas for LNOI-based EOMs and push the development of high-speed PICs.

Funding. National Natural Science Foundation of China (62205129); Natural Science Foundation of Jiangsu Province (BK20200592); Fundamental Research Funds for the Central Universities (JUSRP12024).

Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

REFERENCES

- G. Sinatkas, T. Christopoulos, O. Tsilipakos, and E. E. Kriezis, "Electro-optic modulation in integrated photonics," *J. Appl. Phys.* **130**, 010901 (2021).

Table 1. Comparison of the Proposed EOM with Some Typical Reports on the LNOI-based EOM

Waveguide Structure	Structure Supplement	$V_{\pi}L$ (V · cm)	3 dB Bandwidth (GHz)	L (mm)	Reference
LN ridge	Etched LN [E] ^a	1.75	~40	5	[28]
LN ridge	Etched LN [S] ^b	2.26	~40	11.3	[29]
LN ridge	Etched LN [S]	2.05	~80	10	[30]
LN ridge	Dual-polarization in phase quadrature [E]	2.4	110	23.5	[31]
LN ridge	Dual-capacitor electrode [E]	0.64	~3	4	[32]
LN-Si	Bonded LN on Si [E]	6.7	106	5	[38]
LN-SiN	LN on SiN [E]	6.7	30.6	5	[36]
SiN-LN	Deposited SiN on LN [E]	3	8	12	[37]
Si-LN ridge	Etched LN and deposited Si [S]	1.45	119	4	This work

^aE stands for the experiment.

^bS stands for the simulation.

2. M. Xu and X. Cai, "Advances in integrated ultra-wideband electro-optic modulators," *Opt. Express* **30**, 7253–7274 (2022).
3. K. Liu, C. R. Ye, S. Khan, and V. J. Sorger, "Review and perspective on ultrafast wavelength-size electro-optic modulators," *Laser Photon. Rev.* **9**, 172–194 (2015).
4. M. Lipson, "Compact electro-optic modulators on a silicon chip," *IEEE J. Sel. Top. Quantum Electron.* **12**, 1520–1526 (2006).
5. R. Amin, C. Suer, Z. Ma, I. Sarpkaya, J. B. Khurgin, R. Agarwal, and V. J. Sorger, "Active material, optical mode and cavity impact on nanoscale electro-optic modulation performance," *Nanophotonics* **7**, 455–472 (2017).
6. C. Haffner, D. Chelladurai, Y. Fedoryshyn, A. Josten, B. Baeuerle, W. Heni, T. Watanabe, T. Cui, B. Cheng, S. Saha, D. L. Elder, L. R. Dalton, A. Boltasseva, V. M. Shalae, N. Kinsey, and J. Leuthold, "Low-loss plasmon-assisted electro-optic modulator," *Nature* **556**, 483–486 (2018).
7. C. Wang, M. Zhang, X. Chen, M. Bertrand, A. S. Ansari, S. Chandrasekhar, P. Winzer, and M. Lončar, "Integrated lithium niobate electro-optic modulators operating at CMOS-compatible voltages," *Nature* **562**, 101–104 (2018).
8. P. Pintus, A. Singh, W. Xie, L. Ranzani, M. V. Gustafsson, M. A. Tran, C. Xiang, J. Peters, J. E. Bowers, and M. Soltani, "Ultralow voltage, high-speed, and energy-efficient cryogenic electro-optic modulator," *Optica* **9**, 1176–1182 (2022).
9. X. Fu, J. Cheng, Q. Huang, Y. Hu, W. Xie, M. Tassaert, J. Verbist, K. Ma, J. Zhang, K. Chen, C. Zhang, Y. Shi, J. Bauwelinc, G. Roelkens, L. Liu, and S. He, "5 × 20 Gb/s heterogeneously integrated III-V on silicon electro-absorption modulator array with arrayed waveguide grating multiplexer," *Opt. Express* **23**, 18686–18693 (2015).
10. X. Li, S. Ni, Y. Jiang, J. Li, W. Wang, J. Yuan, D. Li, X. Sun, and Y. Wang, "AlInGaAs multiple quantum well-integrated device with multi-function light emission/detection and electro-optic modulation in the near-infrared range," *ACS Omega* **6**, 8687–8692 (2021).
11. J. Liu, G. Xu, F. Liu, I. Kityk, X. Liu, and Z. Zhen, "Recent advances in polymer electro-optic modulators," *RSC Adv.* **5**, 15784–15794 (2015).
12. F. Qiu and Y. Han, "Electro-optic polymer ring resonator modulators," *Chin. Opt. Lett.* **19**, 041301 (2021).
13. Z. Ma, M. H. Tahersima, S. Khan, and V. J. Sorger, "Two-dimensional material-based mode confinement engineering in electro-optic modulators," *IEEE J. Sel. Top. Quantum Electron.* **23**, 3400208 (2017).
14. B. S. Lee, B. Kim, A. P. Freitas, A. Mohanty, Y. Zhu, G. R. Bhatt, J. Hone, and M. Lipson, "High-performance integrated graphene electro-optic modulator at cryogenic temperature," *Nanophotonics* **10**, 99–104 (2021).
15. Q. Xu, B. Schmidt, S. Pradhan, and M. Lipson, "Micrometre-scale silicon electro-optic modulator," *Nature* **435**, 325–327 (2005).
16. G. T. Reed, G. Mashanovich, F. Y. Gardes, and D. J. Thomson, "Silicon optical modulators," *Nat. Photonics* **4**, 518–526 (2010).
17. L. Chen, Q. Xu, M. G. Wood, and R. M. Reano, "Hybrid silicon and lithium niobate electro-optical ring modulator," *Optica* **1**, 112–118 (2014).
18. E. L. Wooten, K. M. Kissa, A. Yi-Yan, E. J. Murphy, D. A. Lafaw, P. F. Hallemeier, D. Maack, D. V. Attanasio, D. J. Fritz, G. J. McBrien, and D. E. Bossi, "A review of lithium niobate modulators for fiber-optic communications systems," *IEEE Sel. Top. Quantum Electron.* **6**, 69–82 (2000).
19. G. Chen, N. Li, J. D. Ng, H.-L. Lin, Y. Zhou, Y. H. Fu, L. Y. T. Lee, Y. Yu, A.-Q. Liu, and A. J. Danner, "Advances in lithium niobate photonics: development status and perspectives," *Adv. Photon.* **4**, 034003 (2022).
20. Y. Qi and Y. Li, "Integrated lithium niobate photonics," *Nanophotonics* **9**, 1287–1320 (2020).
21. D. Zhu, L. Shao, M. Yu, R. Cheng, B. Desiatov, C. J. Xin, Y. Hu, J. Holzgrafe, S. Ghosh, A. Shams-Ansari, E. Puma, N. Sinclair, C. Reimer, M. Zhang, and M. Lončar, "Integrated photonics on thin-film lithium niobate," *Adv. Opt. Photon.* **13**, 242–352 (2021).
22. J. Lin, F. Bo, Y. Cheng, and J. Xu, "Advances in on-chip photonic devices based on lithium niobate on insulator," *Photon. Res.* **8**, 1910–1936 (2020).
23. Y. Jia, L. Wang, and F. Chen, "Ion-cut lithium niobate on insulator technology: recent advances and perspectives," *Appl. Phys. Rev.* **8**, 011307 (2021).
24. D. Sun, Y. Zhang, D. Wang, W. Song, X. Liu, J. Pang, D. Geng, Y. Sang, and H. Liu, "Microstructure and domain engineering of lithium niobate crystal films for integrated photonic applications," *Light Sci. Appl.* **9**, 197 (2020).
25. M. He, M. Xu, Y. Ren, J. Jian, Z. Ruan, Y. Xu, S. Gao, S. Sun, X. Wen, L. Zhou, L. Liu, C. Guo, H. Chen, S. Yu, L. Liu, and X. Cai, "High-performance hybrid silicon and lithium niobate Mach-Zehnder modulators for 100 Gbit/s⁻¹ and beyond," *Nat. Photonics* **13**, 359–364 (2019).
26. M. Zhang, C. Wang, P. Kharel, D. Zhu, and M. Lončar, "Integrated lithium niobate electro-optic modulators: when performance meets scalability," *Optica* **8**, 652–667 (2021).
27. P. Kharel, C. Reimer, K. Luke, L. He, and M. Zhang, "Breaking voltage-bandwidth limits in integrated lithium niobate modulators using micro-structured electrodes," *Optica* **8**, 357–363 (2021).
28. Y. Liu, H. Li, J. Liu, S. Tan, Q. Lu, and W. Guo, "Low V_π thin-film lithium niobate modulator fabricated with photolithography," *Opt. Express* **29**, 6320–6329 (2021).
29. J. Cai, C. Guo, C. Lu, A. P. T. Lau, P. Chen, and L. Liu, "Design optimization of silicon and lithium niobate hybrid integration traveling-wave Mach-Zehnder modulator," *IEEE Photon. J.* **13**, 2200206 (2021).
30. M. Wang, J. Li, K. Chen, and Z. Hu, "Thin-film lithium niobate electro-optic modulator on a D-shaped fiber," *Opt. Express* **28**, 21464–21473 (2020).
31. M. Xu, Y. Zhu, F. Pittalà, J. Tang, M. He, W. Ng, J. Wang, Z. Ruan, X. Tang, M. Kushnerov, L. Liu, S. Yu, B. Zheng, and X. Cai, "Dual-polarization thin-film lithium niobate in-phase quadrature modulators for terabit-per-second transmission," *Optica* **9**, 61–62 (2022).
32. M. Jin, J. Chen, Y. Sua, P. Kumar, and Y. Huang, "Efficient electro-optical modulation on thin-film lithium niobate," *Opt. Lett.* **46**, 1884–1887 (2021).
33. Z. Yu and X. Sun, "Gigahertz acousto-optic modulation and frequency shifting on etchless lithium niobate integrated platform," *ACS Photon.* **8**, 798–803 (2021).
34. J. Zhang, B. Pan, W. Liu, D. Dai, and Y. Shi, "Ultra-compact electro-optic modulator based on etchless lithium niobate photonic crystal nanobeam cavity," *Opt. Express* **30**, 20839–20846 (2022).
35. P. Zhang, H. Huang, Y. Jiang, X. Han, H. Xiao, A. Frigg, T. G. Nguyen, A. Boes, G. Ren, Y. Su, Y. Tian, and A. Mitchell, "High-speed electro-optic modulator based on silicon nitride loaded lithium niobate on an insulator platform," *Opt. Lett.* **46**, 5986–5989 (2021).
36. N. Boynton, H. Cai, M. Gehl, S. Arterburn, C. Dallo, A. Pomerene, A. Starbuck, D. Hood, D. C. Trotter, T. Friedmann, C. T. DeRose, and A. Lentine, "A heterogeneously integrated silicon photonic/lithium niobate travelling wave electro-optic modulator," *Opt. Express* **28**, 1868–1884 (2020).
37. S. Jin, L. Xu, H. Zhang, and Y. Li, "LiNbO₃ thin-film modulators using silicon nitride surface ridge waveguides," *IEEE Photon. Technol. Lett.* **28**, 736–739 (2016).
38. P. O. Weigel, J. Zhao, K. Fang, H. Al-Rubaye, D. Trotter, D. Hood, J. Mudrick, C. Dallo, A. T. Pomerene, A. L. Starbuck, C. T. DeRose, A. L. Lentine, G. Rebeiz, and S. Mookherjee, "Bonded thin film lithium niobate modulator on a silicon photonics platform exceeding 100 GHz 3-dB electrical modulation bandwidth," *Opt. Express* **26**, 23728–23739 (2018).
39. P. O. Weigel, F. Valdez, J. Zhao, H. Li, and S. Mookherjee, "Design of high-bandwidth, low-voltage and low-loss hybrid lithium niobate electro-optic modulators," *J. Phys. Photon.* **3**, 012001 (2021).
40. COMSOL Multiphysics, "Simulation software," COMSOL Multiphysics (2018), <https://www.comsol.com/comsol-multiphysics> (accessed on 12 October 2022).
41. L. Cai, A. Mahmoud, M. Khan, M. Mahmoud, T. Mukherjee, J. Bain, and G. Piazza, "Acousto-optical modulation of thin film lithium niobate waveguide devices," *Photon. Res.* **7**, 1003–1013 (2019).